Towards Recursive Self Home Improvement

Lifting a weak open model at dishwasher loading with a self-rewriting team

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Abstract

How best to load a dishwasher is a combinatorial, physical problem with an unusually honest reward: a simulated wash decides which items come out clean [8]. We use it as a testbed for self-improving multi-agent systems built only from open, inexpensive models. A deliberately weak loader (Nemotron-3.5-Super-VL, thinking off) is surrounded by a planner, a vision critic, a coach and a librarian, all open Nemotron models on Crusoe serverless inference. Between generations the team rewrites a typed genome of its own prompts, playbook, tool policy and structure, and a deterministic gatekeeper keeps an edit only if it beats the incumbent on paired, fresh tasks. On a held-out exam whose levels contain item kinds and soils never seen in training, the evolved team scores ?? against ?? for the bare loader (?? pp). We report what changed, what it cost, and why the evidence is thin.

1 Introduction

Recursive self-improvement is usually discussed for systems that rewrite their own weights. A humbler, testable version keeps the model fixed and lets the system rewrite the *scaffold* around it: prompts, memory, tools and the division of labour [20, 65, 67]. Such loops are hard to trust: the reward is often an LLM judge, which can be flattered [70], a loop optimising a proxy finds its seams [27, 33, 52], and gains are usually measured where they were selected.

We take the problem of loading a dishwasher, using DishwasherBench, Composo's open (MIT) benchmark [8]. It models one real machine (Bosch SMV4HAX40G): items are placed with deterministic rigid-body physics (Jolt) through 13 MCP tools, then a GPU wash simulation (WebGPU via Dawn) of the Eco 50 programme with one detergent capsule, calibrated on washes of the real machine (parameter set `anchored-v8.owner-capsule`), decides which items are clean against a threshold derived from the IEC 60436 scale [6, 12, 21]. An item scores its points only if it comes out clean. The reward is *objective* (points over points available), *deterministic* (a load always settles and washes the same way, and every episode is verifiable by replay) and *unfakeable*: a bowl mouth-up under a dead spot of the spray arm stays dirty however eloquently the agent describes it. It is also hard: not everything fits, so the player chooses what to load as well as how; jets reach only part of the load [43, 44]; and choosing a subset under capacity is a knapsack over strongly NP-hard 3D packing [24, 34]. Household benchmarks score symbolic goals [45, 49, 51, 55], and prior dishwasher-loading work scores grasp success or agreement with a human preference [22, 23, 30, 56]; to our knowledge none scores whether the items come out clean, the gap DishwasherBench fills.

*Built at the Crusoe × Cerebral Valley Recursive Agents Hackathon, 17 October 2026. Code: `https://github.com/elevenarms/RSHI`.

The benchmark, simulator, calibration, levels and harness are Composo's work [8]. Ours is the self-improving team (*crew*), its memory, Gatekeeper, exam protocol and visualiser, plus cosmetic water, watch and turntable changes to the benchmark's web client that touch neither simulation nor scoring. We claim that a team of cheap open models can lift a weak loader by rewriting the system around it, keeping only changes that measurably help, and that the lift survives on unseen tasks. Contributions:

- A team whose rewritable state is a typed, versioned *genome*, edited only through six typed operations, with the judging roles (memory accounting, acceptance) in deterministic code: the team can change how it plays, never how it is graded.
- A paired acceptance rule over shared seeds in a deterministic simulator, a three-layer memory with shrunken Bayesian confidence, and a held-out exam on task families with new kinds and soils.
- An end-to-end run on open Nemotron models, with lineage, costs and limits.

2 Related Work

Scaffold self-improvement. STOP recursively improves the improver around a fixed LM [65]; ADAS has a meta-agent program new agents [20]; the Darwin Gödel Machine keeps self-edits that validate, in an open-ended archive [67]; Gödel Agent and SICA rewrite their own logic or code [47, 63]. AFlow searches workflows with MCTS and lets cheap models match expensive ones [66]; GPTSwarm optimises prompts and agent connectivity [71]; FunSearch and AlphaEvolve gate LLM proposals with automatic evaluators [36, 48]. Our Coach is a typed, constrained meta-agent in this line.

Prompt and program optimisation. OPRO optimises from a scored history [62]; Promptbreeder and EvoPrompt evolve prompts by LLM mutation [13, 16]; DSPy and MIPRO tune multi-stage LM programs against a metric [25, 40]; TextGrad and Trace propagate textual feedback [7, 64]; GEPA's reflection on traces is closest to our Critic→Coach loop [1].

Memory and multi-agent systems. Reflexion stores verbal reflections [50]; ExpeL and CLIN distil and transfer cross-task insights [32, 68]; Voyager and AWM grow verified skill or workflow libraries [59, 60]; Generative Agents and MemGPT structure retrieval under a context budget [41, 42]. Unlike self-refinement [31], our lessons are counted against physics. Role-based teams [19, 28, 46, 61] and debate [10] follow fixed processes; ours edits its own structure. Voting and verified repeated sampling trade compute for parameters [4, 29, 53]; our verifier is too slow to call per action.

Search over loads. Loading is constrained packing with a physical, black-box objective [58, 69]; genetic algorithms [15, 18], CMA-ES over poses [17], quality-diversity [35], Bayesian optimisation [54] and MCTS over placement sequences [5, 26] are natural per-instance optimisers. We do not search loads di-

Role	Model (open, on Crusoe)	Job
Loader	Nemotron-3.5-Super-VL (no thinking)	plays, 13 tools
Planner	Nemotron-3.5-Super-VL (low effort)	per-item plan
Critic	Nano-Omni-Reasoning-30B (thinking)	post-mortem
Coach	Nemotron-3-Super-120B-A12B	genome edits
Librarian	Nemotron-3-Nano-30B-A3B + code	memory, brief
Gatekeeper	deterministic statistics	accept/reject

Table 1: Roles and models [37–39].

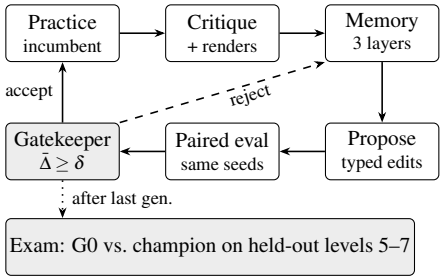


Figure 1: One generation of the loop. Rejections (title, hypothesis, per-task deltas) are logged and shown to the Coach.

rectly: every evaluation costs a wash, and the product is *one load*, not a better loader, so nothing transfers to a new set of items. We optimise the system and test whether its improvements transfer.

3 Method

The team (Table 1, Fig. 1). The Loader plays through the standard DishwasherBench harness tools [8], text only. Optionally a Planner turns the level card and a memory brief into a per-item plan (rack, stance, what to leave out). After each practice episode the Critic gets the item-by-item outcome (rack, stance, clean, residue, retained water), plan, playbook and two renders of the final load, and writes up to five tagged lessons with a polarity. The Coach proposes edits; the Librarian condenses memory into a brief.

Genome and typed edits. A genome is an immutable record of everything the team may rewrite: three prompts (Planner, loader preamble, Critic); a playbook of at most 25 tagged rules with evidence links to lessons; team switches (`usePlanner`, `useMemoryBrief`, `preflightCheck`, which refuses `start_cycle` before a read `check_load`, and the brief's lesson count); and loader settings (reasoning effort, images, denied tools, turn limit). The Coach can change it only by `add_rule`, `edit_rule`, `remove_rule`, `set_prompt`, `set_team` and `set_loader`. A proposal carries a title and a falsifiable hypothesis, should change one coherent thing so gains can be attributed, and is shown past rejections. Invalid edits are dropped; ids hash content and parent, so lineage is diffable. The baseline G0 is the bare loader: no planner, no brief, no preflight, an empty playbook.

Three-layer memory. The *episodic* layer keeps every training episode item by item; the *semantic* layer keeps lessons (from the Critic, and observed from placement statistics), deduplicated on merge, with support s and contradiction c counts; the *procedural* layer keeps every genome and decision. Confidence is a Beta(1,1) posterior mean shrunk towards ignorance while evidence is thin:

$$p = \tfrac{s+1}{n+2}, \quad \kappa = \tfrac{1}{2} + \left(p - \tfrac{1}{2}\right)\tfrac{n}{n+K}, \quad n = s+c,\ K = 3, \qquad (1)$$

so one supporting episode gives 0.54, five of five 0.72, twenty of twenty 0.90. A level's brief lists smoothed clean rates $(k+1)/(m+2)$ of its kinds per rack and stance, then lessons with $\kappa \geq 0.5$ ranked by $\kappa \cdot \text{relevance} \cdot \log(1+n)$. Exam episodes are stored apart and never read back.

Gatekeeper. Candidate C and incumbent I play the same fresh validation drills with the same seeds. Let P be the drills both played validly and x_t the share on drill t; then

$$\bar{\Delta} = \tfrac{1}{|P|}\textstyle\sum_{t \in P}\left(x_t^C - x_t^I\right), \qquad \text{accept} \iff \bar{\Delta} \geq \delta, \qquad (2)$$

and the decision is void if $|P| < 3$. Agent failures (turn or token limit, nothing loaded) score what they scored; *invalid* episodes (provider or harness errors) are replayed on the same task and seed up to twice, and drills still unpaired are dropped for both genomes.[1] Pairing matters because episodes vary enormously: the same baseline genome scored 88% on a Tutorial drill and 0% on a Weeknight drill in one evaluation. Since loading and washing are deterministic, pairing cancels drill difficulty, leaving the genome's effect plus sampling noise. The accepted candidate with the largest $\bar{\Delta}$ becomes the incumbent; its validation episodes are the next practice set.

Exam: what "held-out" means. Training and validation drills come from DishwasherBench course levels 1–4 (tutorial, lunch for four, weeknight dinner, breakfast), the exam from levels 5–7 (mixed family, baking day, dinner party) [8]. These are new task families, not just new seeds: they hold kinds and soils training never showed (baby bottle and teat, spouted beakers, lunch tubs, baking tray, tin, platter, board; burnt sugar, overnight soils). G0 and the champion play identical exam drills; nothing is learned from them.

4 Experimental Setup

All models run on Crusoe Managed (serverless) Inference via an OpenAI-compatible API, with a client-side rate limiter and byte-stable prompts for prefix caching (83–90% of input tokens hit the cache in smoke runs). In a sweep (40-request bursts, 60 s at 8 in flight) only Nemotron-3.5-Lightning and Gemma-4-31B returned HTTP 429s; the Nano models sustained ∼300 calls/min at 370 ms median.

Choosing a weak loader. We want headroom: a loader that saturates leaves the Gatekeeper nothing to measure. Table 2 shows one smoke per model on the same drill. Nemotron-3.5-Super-VL with thinking off is the weakest model that still plays: everything it loads comes out clean, but it runs out of turns. Without thinking the Nano models loop; with it, Nano-30B plays but takes ∼6× Super-VL's wall time, too slow for an evolution loop. Our pilot used Nemotron-3.5-Lightning-30B-A3B (41–70% in smokes) until its shared deployment throttled

[1] Our first version counted provider errors as agent failures. In a pilot, generation 2's fast, non-thinking loader hit a shared deployment's request cap (HTTP 429) and both decisions were voided. On cheap shared endpoints rate limits are a real systems constraint, and an acceptance rule must not confuse infrastructure noise with policy quality.

Loader (thinking)	Share	Loaded/clean	Turns	Time
DeepSeek-V4.1-Flash (off)	100%	12/12	21	90 s
Qwen3.8-27B (off)	100%	12/12	19	199 s
GLM-5.3-Flash (low)	93%	10/10	40	97 s
DeepSeek-V4-Flash (off)	89%	12/11	40	89 s
gpt-oss-120b (low)	49%	8/7	40	154 s
Nemotron-3-Nano-30B (low)	41%	5/5	40	870 s
Nemotron-3.5-Super-VL (off)	**21%**	**3/3**	**40**	**129 s**
Nemotron-3-Nano-30B (off)	0%	1/0	loops	
Nemotron-3-Nano-Omni (off)	0%	2/0	loops	

Table 2: Choosing the loader: one smoke per model on the same 12-item Breakfast drill (seed 7, 61 points, 40-turn cap). We take the weakest that still plays.

Exam drill	Source	Avail.	G0	Champ.	Δ
Family #0	mixed-family	59	??	??	??
Baking #1	baking-day	71	??	??	??
Party #2	dinner-party	77	??	??	??
Family #3	mixed-family	71	??	??	??
Total points		278	??	??	??
Mean share			??	??	??

Table 3: Held-out exam: points per 12-item drill for G0 and the champion **G?-??????**, same drills and seeds, fast tier.

us to about five calls per minute. In that pilot's generation 1 the Gatekeeper accepted "Enable a Planner to guide loading" at $\bar{\Delta} = +0.135$ and rejected a bowl-mouth-down playbook rule at -0.095: structure helped, a plausible domain rule did not.

Protocol. A drill is 12 items of a DishwasherBench level drawn by seed, cutlery in proportion. The final run (`2026-10-17T20-49-49-evolve`) has 3 generations: 4 baseline practice drills, 2 proposals and 3 paired fresh validation drills per generation, $\delta = 0.03$, a 40-turn cap per episode, 8 game servers with one wash queue each and 8 episodes in flight, then an exam on 4 held-out drills (Table 3). Washes use the fast tier (4 mm blocks rather than 2 mm), whose verdicts agree with the official tier's on 91% of the standard load and 97% of pieces on the course levels [8].

Baseline. G0 scored 39% on its practice drills: Breakfast 56% (8 of 8 loaded items clean), Tutorial 44% (8 of 8, wash started early with 4 items left out), Weeknight 32% (5 loaded, 3 clean), Lunch 24% (2 of 2). It fails at committing to placements within the turn budget, not at knowing what washes clean, which suggests where a team can help.

5 Results

Exam. The champion (**G?-??????**, generation **??**) scores a mean share of **??** against **??** for G0 (**??** pp), ahead on **??** drills (Table 3). It loads **??** items against **??**, with clean-of-loaded rates of **??** and **??**; failed episodes **??** vs. **??**. Since share = loaded fraction × clean rate, the baseline's failure mode predicts most of any gain in the first factor.

Evolution. The Gatekeeper accepted **??** of **??** candidates over 3 generations (Fig. 2); memory ended with **??** lessons from **??** episodes. In generation 1 the incumbent scored 88%, 15% and 0% on Tutorial, Lunch and Weeknight (mean 34.2%). G1-579472 (**"title?"**) scored 69%, 40%, 0% (36.3%, +2.1 pp): rejected (below the margin). G1-bc908e (**"title?"**) scored 72%, 42%, **??** (**??**, **??** pp): **accepted/rejected?**. Accepted edits:

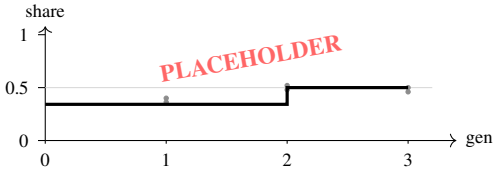


Figure 2: Incumbent (steps) and candidates (dots), mean validation share. Seeds are fresh each generation: only within-generation differences compare.

- **G?: "title"** (`op ...`), $\bar{\Delta} = +?.??$ **(or: none accepted).**

Cost. The run took **??** and **??** input tokens (**??** cached) plus **??** output tokens, an inference cost of **??**.

6 Discussion and Limitations

The design point we would defend is the division of authority: models propose anything the edit grammar allows, but physics scores it and code decides, which LLM-judged loops lack [2, 70]. The open question is credit assignment between structure (a planner that commits the loader to a list) and content (lessons about what washes clean).

Limitations. *One machine*: one dishwasher, programme and detergent; the exam tests transfer to new items, soils and loads, not new physics. *Small n*: 3 validation and 4 exam drills, one exam pass; no confidence interval worth the name, and a paired permutation or sign test [9] on so few drills has little power. *Stochastic sampling*: the simulator is deterministic but the models are not, so pairing removes task variance, not policy variance. *Fast tier*: a few percent of verdicts differ from the official tier. *Multiple comparisons*: 2 candidates per generation are each tested at a fixed margin and the best is kept, so validation gains are biased upwards, and repeated accept/reject on one benchmark overfits it [3, 11, 14]; the held-out exam, not the validation curve, is the evidence. *Margin*: $\delta = 0.03$ over 3 drills is about one average item (some 70 points over 12 items) on a single drill, far below the run-to-run noise of a sampling loader; a sequential test [57] or repeated validation would be the principled fix.

7 Conclusion

A team of open, inexpensive models can rewrite the system around a weak loader, keep only what the physics confirms, and carry the improvement to tasks with objects it has never handled. The dishwasher is a small domain, but its reward cannot be argued with, which is exactly what recursive self-(home-)improvement needs. Next: more exam passes, an official-tier exam, and a placement-search tool the team may choose to use.

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